A3-PS003-TC137

ENSO — Federated Traffic-Flow Prediction & Communication-Cost Evaluation

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Problem Statement 03 (PS003)

Problem Description

Three connected bottlenecks in traffic-aware, low-waste logistics

01 Bandwidth overload

Continuous raw GPS uploads create avoidable network traffic and recurring communication costs.

02 Privacy exposure

Centralizing detailed movement traces can reveal driver locations, routes and commercial activity.

03 Empty-mile inefficiency

Unused vehicle capacity and fragmented local demand data make it harder to coordinate compatible loads.

Core challenge: predict local traffic patterns and coordinate spare capacity without routinely transferring raw location histories.

The 30–40% empty-running figure is a motivating estimate from the initial brief, not a verified ENSO baseline.

Background

Why federated learning fits a distributed traffic network

Traffic data is distributed

Vehicles observe different roads, times, congestion levels and capacity conditions.

Centralization has trade-offs

A central model can learn globally, but raw-data transfer adds cost, latency and privacy risk.

Federated learning

Each node trains locally; model updates—not raw trip histories—are aggregated centrally.

Logistics coordination

Traffic forecasts can inform route timing and identify opportunities to match nearby goods with spare capacity.

Design principle: keep sensitive data local by default, transmit only what the learning task requires, and measure the communication/accuracy trade-off.

Literature & RFC Review

Relevant foundations to inform the implementation

Federated Averaging (FedAvg)

McMahan et al. (2017): a foundational approach for aggregating locally trained model updates.

Secure Aggregation

Bonawitz et al. (2017): techniques that help a server aggregate updates without inspecting each participant’s update in plaintext.

Traffic prediction research

Traffic forecasting commonly uses temporal and spatial patterns; ENSO must compare its forecasts against a simple baseline.

Standards / interfaces

Use documented API contracts, TLS in transit, authenticated clients and explicit model-versioning for update exchange.

Gap addressed: evaluate prediction quality and bytes transferred together, rather than reporting model accuracy alone.

Proposed Solution

ENSO combines edge learning with capacity-aware logistics

Local learning

Train a lightweight traffic model on locally available speed, time and route features.

Private update exchange

Send versioned model updates; avoid uploading raw GPS histories by default.

Global coordination

Aggregate updates, publish a global model and expose route/capacity recommendations.

Logistics layer

Use predicted congestion and available vehicle space to suggest compatible local produce pickups along existing journeys. Drivers or operators confirm matches; ENSO does not assume every spare space is suitable.

Privacy note: federated learning reduces raw-data sharing, but model updates can still leak information without safeguards.

Optimization Proposed

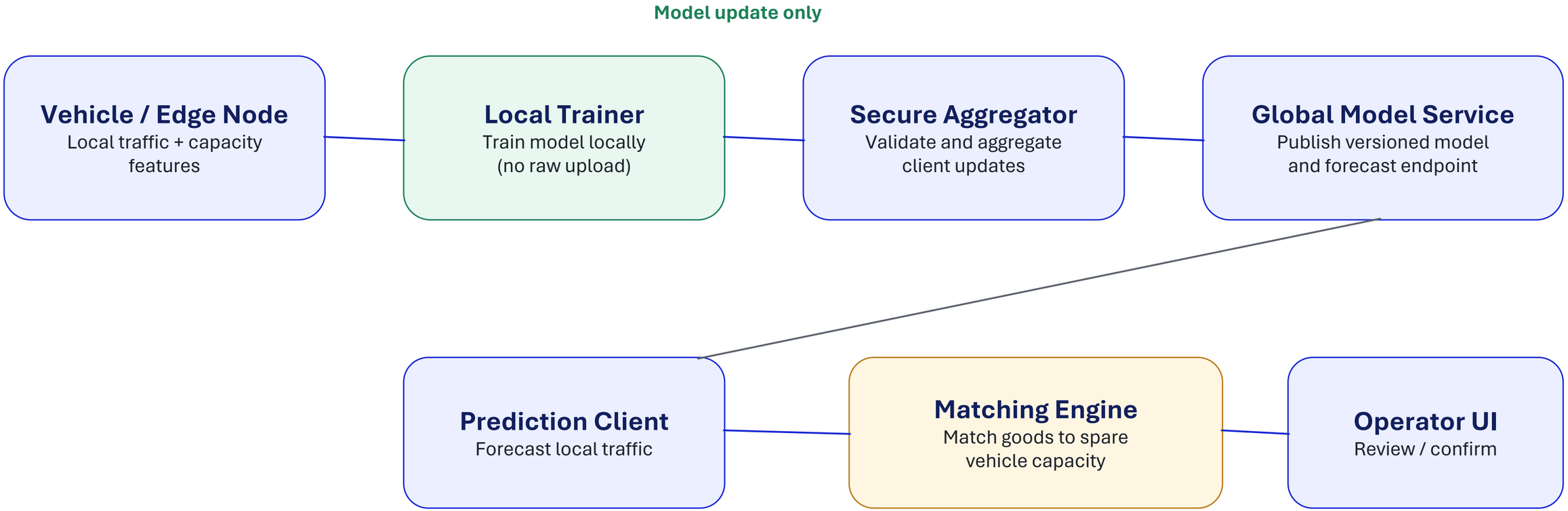
Compare the baseline with ENSO; validate each improvement experimentally

Dimension	Centralized baseline	ENSO target / evaluation
Data movement	Frequent raw GPS uploads	Exchange model updates; measure bytes per round
Privacy design	Raw traces collected centrally	Keep raw traces local; test secure aggregation
Communication cost	Baseline measured in bytes/vehicle/day	Target: materially lower payload; 80% is a hypothesis to test

Report both model quality (e.g., MAE/RMSE) and communication (bytes per client, round and day). Include failed rounds and reconnect overhead.

Solution Architecture & Design

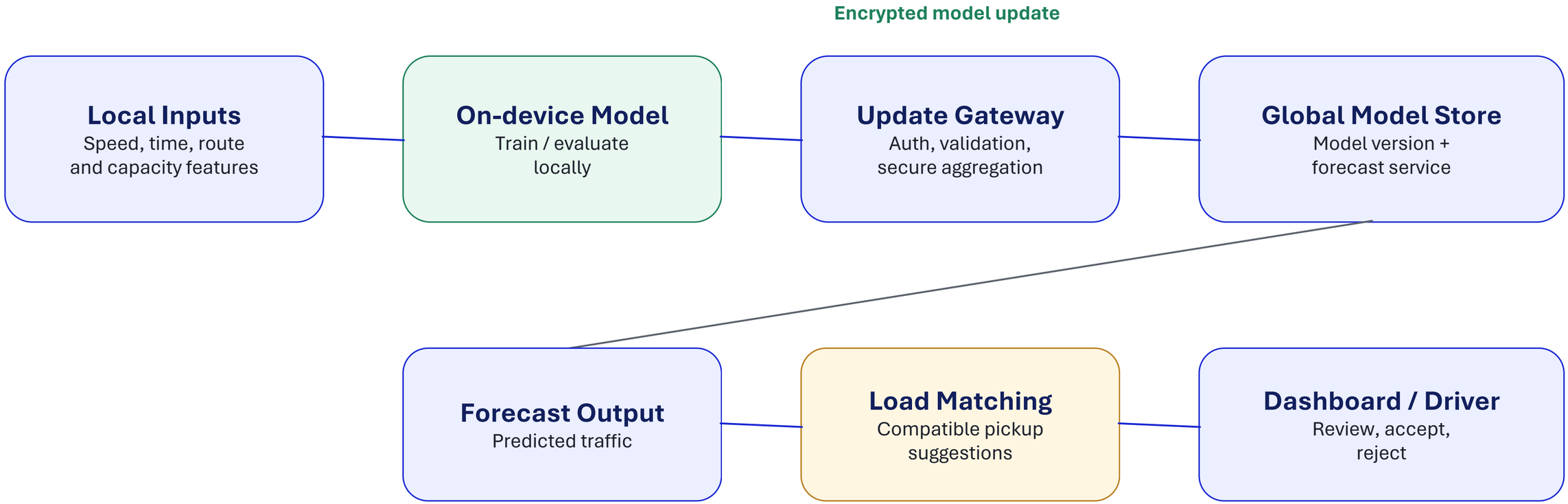
Federated training and route/capacity recommendation loop



Trust boundaries: device data stays local; authenticated encrypted updates flow to aggregation; forecasts and recommendations flow back to authorized clients.

Data Flow Diagram (DFD)

What moves through the system—and what stays on device



Not transmitted by default: raw GPS history and detailed local trip records. Implementation should log payload size and avoid logging sensitive fields.

Deployment Timeline

A staged proof-of-concept with measurable gates

1. Define baseline	**Weeks 1–2**	Select traffic dataset, define forecast horizon, baseline model and communication metrics.
2. Build edge prototype	**Weeks 3–4**	Train locally; implement update serialization, authentication and model versioning.
3. Aggregate & secure	**Weeks 5–6**	Add aggregation, transport encryption, access controls and failure handling.
4. Evaluate & demo	**Weeks 7–8**	Compare accuracy, bytes transferred, latency and load-match acceptance in controlled tests.

Gate to proceed: show lower communication volume without unacceptable loss of prediction quality.

Bibliography

Foundational references and implementation documentation

- [1] H. B. McMahan et al. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” AISTATS, 2017.
- [2] K. Bonawitz et al. “Practical Secure Aggregation for Privacy-Preserving Machine Learning.” ACM CCS, 2017.
- [3] TensorFlow Federated. Official documentation: https://www.tensorflow.org/federated
- [4] PyTorch. Official documentation: https://pytorch.org/docs/stable/
- [5] ENSO evaluation plan: compare centralized and federated baselines using the same data split, prediction horizon and measurement window.

Thank You for Your Attention!

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